

Topological Segmentation Error and Boundary-Condition Tuning in Computed Coronary FFR: A Controlled In Silico Study

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Abstract—Fractional flow reserve, a pressure ratio guiding coronary revascularization, can be computed from computed tomography angiography. The computation depends on the segmented artery and on outlet boundary conditions derived from it, often tuned to measured perfusion, as in a digital twin. We tested in silico whether a perfusion-matched model can be misclassified at 0.80. We inserted 150 stenoses into 108 coronary trees, applied four segmentation errors (missed side branch, vessel break, longer lesion, narrowed taper), and recomputed fractional flow reserve with a reduced-order model under fixed, re-derived and tuned boundary conditions, with one global or per-territory parameter fitted to clean-model territory flows, in discrete and leaky (distributed-outflow) microvascular beds. With fixed boundary conditions, branching (topological) errors changed the decision in 32–33% of models, against 6–10% for vessel-size (caliber) errors of inter-observer magnitude. Re-derived or tuned boundary conditions reduced topological decision changes to 5–19% (pooled). In this threshold-stratified cohort, tuned models still passed a perfusion check stricter than measurement repeatability while wrong by more than 0.05: 19–20% (discrete bed) and 5–8% (leaky bed) of topological-error models and 4–18% of caliber-error models, mostly taper, against 3–4% for correct anatomy and none for caliber errors with re-derived boundary conditions. Per-territory tuning corrected the typical missed branch; in one three-dimensional case it reduced a shift of 0.074 to -0.0007 . The directions held at doubled and, in the leaky bed, tripled demand. A perfusion match after tuning does not show that the computed value is correct; branching and caliber should be checked before tuning.

Index Terms—Boundary conditions, coronary artery disease, digital twins, fractional flow reserve, image segmentation, reduced-order modeling, uncertainty.

I. INTRODUCTION

FRACTIONAL flow reserve (FFR) is a pressure ratio that guides the treatment of a coronary narrowing. It is the mean blood pressure beyond the narrowing divided by the mean aortic pressure during maximal flow. A value at or below 0.80 marks a stenosis for revascularization [1]. FFR can also be computed without a pressure wire. The

lumen, the blood-filled channel of the artery, is segmented from coronary computed tomography (CT) angiography and converted into a flow model, and specific implementations of the computed FFR have been validated against invasive measurement in a multicenter trial [2]. The calculation exists in three-dimensional (3D), reduced-order and machine-learning forms [3]–[6].

Every such model needs boundary conditions, the pressures, flows or resistances imposed where the model ends. They stand in mainly for the microvascular bed, the arterioles and capillaries that set perfusion, the blood flow to each region of heart muscle. The bed resistance is not measured but derived from the segmented lumen through scaling laws that relate vessel size to the flow it carries [4], [7]. It is increasingly tuned so that the model reproduces measured perfusion [8] or measured flow splits [9]. In a coronary digital twin, this tuning personalizes the physiology. The segmentation therefore enters the computed FFR twice, through the arteries and through the boundary conditions derived from them.

Segmentation error has been studied mainly as caliber error. Caliber is the inner diameter of a vessel, so a caliber error is a lumen drawn too wide or too narrow, or a narrowing drawn too long, while the branching is unchanged. Sensitivity analyses perturb the radius at fixed bifurcations and report a continuous change in the computed pressure index [10]–[13]. Studies that set geometry against boundary conditions rank the two by their influence on a continuous output [14]–[16]. Topological error, a change in the branching structure itself such as a missed side branch or a vessel that ends too early, has received less attention. In a pulmonary network model, connectivity contributed more to hemodynamic uncertainty than vessel radius and length [17]. In a recent coronary benchmark, every evaluated method produced vessel breaks despite high overlap scores [18], and small distal segments remain the hardest to segment [19]. A missed branch removes part of the microvascular bed from the model and changes both the flow through the stenosis and the boundary conditions derived from the tree.

Whether such an error reaches the computed FFR depends on how the boundary conditions are handled. With outlet resistance held fixed, adding a side branch beyond a stenosis lowered FFR by 15.5% in an idealized model and by 13% in one of two models built from optical coherence tomography [20]. When a cohort-wide outlet resistance was recalibrated for a model with side-branch outflow, diagnostic accuracy changed little [21]. In a clinical cohort, a change of outlet model moved

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sensitivity from 58.1% to 68.6% at an unchanged area under the curve [5]. If the boundary conditions are tuned until the model matches measured perfusion, a topological error may vanish from the perfusion check yet remain in the pressure along the vessel. None of these studies compares fixed, re-derived (derived anew from the segmented tree) and tuned boundary conditions on the same corrupted anatomy at 0.80, or tests whether a tuned model can pass a perfusion check while its FFR is wrong.

We propose a controlled test for this problem, an ablation in which one segmentation error at a time is introduced into an otherwise correct model, whose FFR serves as the reference value, so every change in FFR can be traced to the error and to the handling of the boundary conditions. The approach is new in three respects: it judges segmentation error by the treatment decision at 0.80 against the variability of repeat invasive FFR; it solves the same corrupted anatomy under fixed, re-derived and tuned boundary conditions, with one global or one per-territory tuning parameter and in two microvascular bed structures, and separates topological from caliber errors of inter-observer magnitude; and it counts how often a model passes a perfusion check while its FFR is materially wrong, with a 3D computational fluid dynamics (CFD) case as a check.

II. METHODS

A. Data and Cohort

ImageCAS-X provides voxel-wise lumen masks, labeled centerlines and surfaces for 800 CT angiography scans of the public, anonymized ImageCAS dataset [18], [22], [23]. We used its 160-scan test split, in which a second analyst re-annotated each scan. The two annotators agreed with a Dice similarity coefficient (DSC, the overlap of two masks) of 92.8% and a 95th-percentile Hausdorff distance (HD95, the surface distance exceeded by 5% of boundary points) of 2.46 mm [18]. Each scan gives a left and a right coronary tree. The radius at each centerline point is the Euclidean distance from the point to the nearest background voxel of the mask.

The dataset contains few lesions near the decision threshold, so we inserted one stenosis per instance into an otherwise unchanged tree. An instance is one lesion in one tree. Its clean model is the lesioned tree with no segmentation error. A host vessel (left anterior descending, LAD; left circumflex, LCx; or right coronary artery, RCA) was eligible when image quality was at least adequate (2 on a 0–4 scale), the taper-fit radius r_{fit} (a linear fit of radius along the vessel) at the lesion center was at least 1.0 mm, native narrowing relative to r_{fit} was below 40% diameter stenosis (DS), the measurement point 20 mm beyond the lesion lay on a vessel of radius at least 0.75 mm, and the FFR before insertion was at least 0.90. The lesion is an axisymmetric cosine narrowing (Supplementary Material),

$$r'(s) = \min(r, (1 - w)r + w r_{\text{fit}}(1 - \text{DS})), \quad (1)$$

$$w = \frac{1}{2} \left[1 + \cos \frac{2\pi(s-c)}{L} \right]$$

for $|s - c| < L/2$ ($r' = r$ elsewhere; DS as a fraction), where s is arc length, c the lesion center and L its length. The vessel was never widened. Lesions were centered 20 or 45 mm from

the vessel origin, were 10 or 20 mm long and had 40, 50, 55, 60, 65, 70, 75 or 80% DS. Of the 6944 eligible instances (280 host vessels, 140 patients), we drew 25, at random with a fixed seed, in each of six 0.05-wide bands of leaky-bed baseline (clean-model) FFR from 0.65 to 0.95, at most two per tree, giving 150 instances (50 each in the LAD, LCx and RCA) in 108 trees from 93 patients (Supplementary Material).

B. Reduced-Order FFR Model

All 150 instances were solved with a reduced-order model, which represents the coronary tree as a network of flow resistances instead of computing the full 3D flow field. The model places one node at every centerline point. Each element between consecutive nodes has the Poiseuille resistance $8\mu \Delta s / (\pi r^4)$ of its actual radius r and length Δs , with blood viscosity $\mu = 0.004$ Pa s. Where the narrowing exceeds 30% DS, an expansion loss $\Delta p = K Q |Q|$ is lumped at the stenosis, with $K = \rho K_t (A_0/A_s - 1)^2 / (2A_0^2)$, $K_t = 1.52$ [24], density $\rho = 1060$ kg m⁻³, and A_0 and A_s the reference and throat areas. Aortic pressure is 90 mmHg and venous pressure 5 mmHg. The nonlinear equations are solved by under-relaxed fixed-point iteration, each step a sparse linear solve for the node pressures, until the relative flow change falls below 10^{-8} . The mass-conservation error is recorded for every solve. FFR is the computed pressure at the measurement point divided by aortic pressure. The model is steady and represents maximal hyperemia (maximal flow) without autoregulation.

The microvascular bed is modeled as conductances (the inverse of resistance) that drain flow from network nodes to venous pressure. Each conductance depends on a healthy reference radius r_{ref} (the taper-fit radius r_{fit} made non-increasing along each path) and is scaled by one bed constant C_b . Two bed structures were analyzed because they treat a deleted vessel differently. In the leaky bed, flow leaves along the vessel wall at every node of the tree truncated at $r_{\text{ref}} = 0.50$ mm, near the mask resolution, with conductance $(r_{\text{ref}}^3 - \sum r_{\text{ref,child}}^3) / C_b$, so that flow follows Murray's cube law [7]. This distributed outflow stands for the side branches the image does not resolve, and the conductance of a deleted branch reappears as wall outflow at its parent node. In the discrete bed, flow leaves only at the outlets of the tree truncated at $r_{\text{ref}} = 0.60$ mm, the cut that kept a side branch beyond most lesions, with conductance $r_{\text{ref}}^{2.66} / C_b$. The exponent is close to 8/3, the inverse of the 3/8 power that relates coronary vessel diameter to the myocardial mass it perfuses [25], so outlet flow is proportional to perfused mass. The discrete bed is the outlet structure that a 3D model can represent. In both structures C_b was calibrated on the healthy-equivalent network, the tree with its reference radius and no lesion, so that inflow equals the hyperemic demand $Q = k r_{\text{in}}^3$, where r_{in} is the inlet radius. With the model setting $k = 562$ s⁻¹, a vessel with the normal proximal LAD diameter of 3.7 mm [26] carries 214 mL/min, within one standard deviation of the hyperemic flow measured in that artery by continuous thermodilution (228 ± 71 to 293 ± 102 mL/min) [27]. The segmented arteries were narrower than these normal values (median inlet radius 1.60 mm), so the cohort's median demand was 2.28 mL/s (137 mL/min)

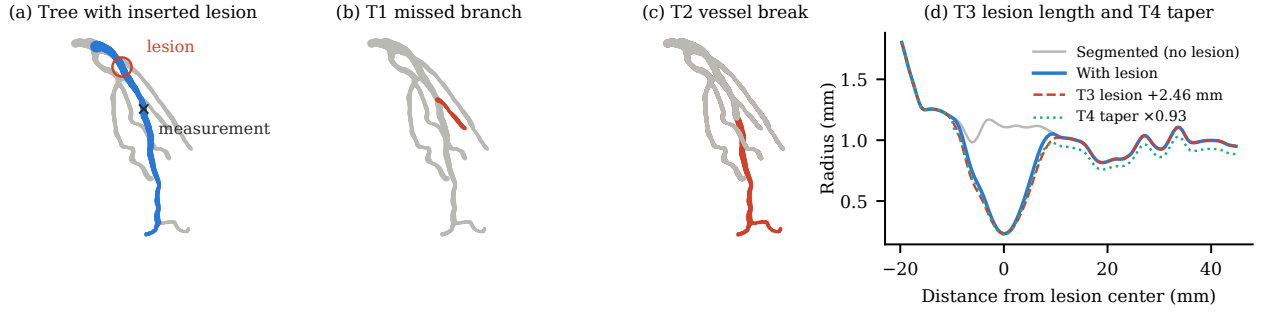


Fig. 1. The four segmentation error types on one cohort tree (left coronary tree of scan 14, front view, line width proportional to radius). (a) The tree with the host LAD in blue, an inserted 80% DS proximal lesion and the measurement point 20 mm beyond it. (b) T1 deletes the largest side branch beyond the lesion (red). (c) T2 ends the host vessel 25 mm beyond the distal edge of the lesion; the red part is lost. (d) Radius along the LAD for the caliber errors, T3 lengthening the lesion by 2.46 mm and T4 shrinking the radius by 7% from the lesion onwards. This instance was also solved in 3D without the lesion, with the lesion, and with T1.

per tree, and the full pipeline was repeated with k doubled and tripled. Reference radii and C_b are computed from the original tree, so an inserted lesion changes only the epicardial radius. The leaky bed was analyzed on all 150 instances. The discrete bed was analyzed on the 97 instances whose healthy-equivalent network gave a main-vessel FFR of at least 0.90 and had at least two outlets, and baseline FFR bands were derived separately for each bed. Halving the centerline spacing changed FFR by at most 0.005 in either bed.

C. Segmentation Error Types

Four error types were applied to each instance with its lesion in place (Fig. 1). Two are topological errors, which change which vessels exist. Two are caliber errors, which change the lumen size or lesion extent while the branching is unchanged. The missed branch (T1) deletes the largest side branch that leaves the host vessel beyond the lesion, together with its subtree, and is possible only where such a branch exists. The vessel break (T2) ends the host vessel 25 mm beyond the distal edge of the lesion, so the measurement point survives but the run-off beyond it is lost. The lesion length (T3) re-inserts the same stenosis 2.46 mm longer about the same center. The taper (T4), a uniform narrowing, multiplies the radius by 0.930 from the proximal edge of the lesion onwards and in every descendant vessel. For T3 and T4 the set of modeled nodes is fixed to that of the clean tree, so a narrowed radius does not remove an outlet.

The caliber magnitudes are the inter-observer statistics of the test split, so each caliber error is as large as the disagreement between two annotators. HD95 (2.46 mm), which bounds the disagreement in lesion extent, sets T3. The DSC of 92.8% sets T4 as the radius ratio λ of two coaxial cylinders, $2\lambda^2/(\lambda^2 + 1) = 0.928$. Because both annotators edited the same automatically generated centerlines, the dataset authors describe the inter-observer DSC as an upper bound on agreement [18]. T1 and T2 have no measured magnitude and are design choices, so the topological results are conditional on them. Nodes of the corrupted tree were matched to the clean tree by coordinate, and FFR was read at the same point.

D. Boundary-Condition Protocols

Each corrupted tree was solved under four protocols. Under Protocol A (fixed), surviving nodes keep the bed conductances and C_b of the clean model, and the bed of deleted vessels is lost. This represents the case in which the boundary conditions were obtained correctly despite the segmentation error. Under Protocol B (re-derived), the bed rule is applied to the corrupted tree and C_b is recalibrated to that tree's own hyperemic demand, as an automated pipeline does with whatever anatomy it receives. Under Protocol C (tuned), one global scaling of the re-derived C_b is fitted so that the corrupted model reproduces the clean model's territory flows, which stand in for an error-free perfusion measurement. A perfusion territory is the subtree below each child of the first bifurcation of the clean tree. Its target is the total bed outflow of that subtree in the clean model, including the share of any vessel the error deleted, because the heart muscle is perfused whether or not its artery was segmented. Each corrupted node belongs to the territory of its clean counterpart. The fit minimized the mean squared relative error of the territory flows on a 61-point grid over ± 1.5 decades of C_b , then refined it; a fit at the edge of the range was a failure (none occurred). One parameter against two or more targets makes the fit over-determined, so this is the least flexible tuning tested. Under Protocol D (per-territory tuned), the re-derived bed of each territory is scaled by its own factor, fitted (within $10^{\pm 3}$) so that every territory flow equals its clean target. With one parameter per target the fit is exactly determined, the most flexible tuning to territory flows. Protocols C and D are defined only when at least two territories retain a vessel. They were undefined for 33 of 77 (discrete) and 47 of 118 (leaky) missed-branch instances and for 36 of 96 and 49 of 149 vessel breaks, all but four in the RCA. Both protocols match flow only, because matching both flow and pressure at the outlets of a steady tree would reproduce the clean FFR by construction.

For every protocol we computed the perfusion residual, the root mean square of the relative differences between the model's territory flows and the clean targets. It is the mismatch that a perfusion check reports.

E. Three-Dimensional CFD Setup

To test whether the findings depend on the simplifications of the reduced-order model, one instance (a 20 mm lesion of 80% DS in the proximal LAD) was also solved with 3D CFD, which computes the full velocity and pressure fields from the Navier–Stokes equations by the finite-volume method. Three lumens were built: without the lesion, with the lesion (baseline), and with the lesion and the missed branch (T1). Surfaces were extracted from the mask by marching cubes and Taubin smoothing; the lesion was applied by moving surface vertices radially with (1), and the branch was removed by deleting its voxels. Meshes of 3.5–3.9 million cells (cfMesh) had 25 μm cells within ± 4 mm of the throat and four boundary layers.

Steady laminar Newtonian flow, with the viscosity and density of the reduced-order model, was solved with OpenFOAM (simpleFoam, ESI v2406; SIMPLE coupling; bounded second-order upwind convection). Each solve ran 3 000 iterations; convergence required scaled residuals below 10^{-5} and a steady pressure at the measurement point. The walls were rigid with no slip, and the inlet carried aortic pressure. Each outlet either set its pressure to venous pressure plus the clean tree's resistance times its flow (Protocol A), or received a prescribed clean territory flow, shared over the territory's surviving outlets by bed weight (the per-outlet limit of Protocol D). FFR is the area-averaged pressure on a cross-section at the measurement point divided by aortic pressure. A grid-convergence study on an idealized 70% DS stenosis (throat cells of 50, 25 and 12.5 μm) gave a discretization uncertainty in FFR of 5.5×10^{-4} . A reduced-order counterpart of each geometry, rebuilt on the area-equivalent radius of the meshed lumen with the same outlet conditions, separates the effect of model fidelity from that of radius definition.

F. Outcomes and Statistical Analysis

For each instance, error type, protocol and bed, ΔFFR is the corrupted model's FFR minus that of the instance's own clean model in the same bed. A flip is a change of classification at 0.80 between the two, that is, a change in the indicated treatment. A model passes the perfusion check when its perfusion residual is below 10%. Published repeatability of hyperemic myocardial blood flow (within-subject variation of 10–15% [28] and a regional repeatability coefficient of 30–37% [29]) implies a 95% bound of about $1.96 \times (10\text{--}15\%) \approx 20\text{--}29\%$ for one model output against one measurement, so the 10% threshold is stricter than measurement repeatability. Thresholds of 13% and 16% were analyzed as sensitivities. An FFR is materially wrong when $|\Delta\text{FFR}| > 0.05$, half the width of the 0.75–0.85 grey zone [30].

Proportions are reported with Wilson 95% confidence intervals (CIs). Protocol contrasts of flips (B against A, C against A, C against B) are paired within instance and tested with the exact McNemar test, with Holm adjustment across the three contrasts of each error type and bed; passes-and-wrong (models passing the check while materially wrong) under tuning (C and D) are compared with Protocol B by the same test. Topological and caliber errors are compared

by a paired sign test on the per-instance flip proportions of each class, and paired changes in $|\Delta\text{FFR}|$ by the Wilcoxon signed-rank test. Instances are clustered within patients, so the confidence intervals are descriptive. A Bayesian mixed-effects logistic model of flips with patient and lesion-slot (host vessel, position and length) random effects agreed with the paired tests (Supplementary Material). Every analysis was run in both beds, effect sizes are reported per bed, and a finding is claimed only where its direction agrees in both.

Each flip rate is compared with a noise floor, the rate expected from measurement variability alone. For each instance, this is the probability that a repeat invasive FFR falls on the other side of 0.80, following the measurement-certainty approach of [30]: the repeat is normal about the clean value, taken as the first measurement, with the published standard deviation of the test–retest difference, 0.018 [31]. A second floor was simulated on the clean anatomy: 20 draws per instance perturbed cardiac output, aortic pressure, blood viscosity, the share of flow to each territory and the measured territory flows by their within-subject variability, and the model was tuned to the perturbed flows as in Protocol C.

Because the cohort was stratified to span 0.80, its rates are conditional on that design and are not clinical prevalences. The analysis pipeline is shown in the Supplementary Material.

III. RESULTS

A. Decision Changes by Error Type and Protocol

Topological errors changed the decision more often than caliber errors under Protocols A–C in both beds (Table I). With fixed boundary conditions in the discrete bed, 45 of 137 topological-error models (33%, 95% CI 26–41%) crossed 0.80, against 20 of 194 caliber-error models (10%, 7–15%). In the leaky bed the proportions were 32% (26–38%) and 6% (4–9%) (paired sign test, $p \leq 0.002$ in both beds). Under Protocols B and C the difference kept its direction. It was significant in the discrete bed ($p = 0.013$ and 0.002) but not in the leaky bed ($p = 0.18$ and 0.057). Under Protocol D the two classes flipped at similar rates (8% and 11% discrete, 6% and 4% leaky). Flips were concentrated in the bands adjacent to 0.80 (Fig. 2). All 2 599 corrupted-model solves under Protocols A–C and 769 under Protocol D converged (maximum mass-conservation error 9×10^{-9}).

The caliber errors rarely changed the decision. At the magnitude of inter-observer disagreement they changed FFR by a mean of 0.006–0.037 in absolute value, and their flip rates of 1–14% were of the order of the 5.0–6.5% expected from repeat invasive measurement. Only the taper exceeded this noise floor with its lower confidence bound, in the discrete bed under fixed boundary conditions (13%) and Protocol D (14%). Against the grey zone of 0.75–0.85 [30], no caliber-error flip under fixed boundary conditions ended outside the zone, whereas 20% (discrete) and 23% (leaky) of topological-error models did (Supplementary Material).

Of Protocols A–C, re-deriving the boundary conditions (Protocol B) produced the fewest flips. Against Protocol A it reversed 14–56 topological flips per cell and created none (McNemar, Holm-adjusted $p < 0.001$), except for discrete-bed

TABLE I
DECISION CHANGES AND PASSES-AND-WRONG BY ERROR TYPE, PROTOCOL AND BED STRUCTURE

Error	Protocol	Discrete bed (97 instances)			Leaky bed (150 instances)		
		<i>n</i>	Flip, %	Passes and wrong, %	<i>n</i>	Flip, %	Passes and wrong, %
T1	A	77	27 (19–38)	13 (7–22)	118	18 (12–26)	16 (11–24)
	B	77	9 (4–18)	1 (0–7)	118	5 (2–11)	5 (2–11)
	C	44	18 (10–32)	20 (11–35)	71	8 (4–17)	11 (6–21)
	D	44	2 (0–12)	23 (13–37)	71	6 (2–14)	4 (1–12)
T2	A	60	40 (29–53)	12 (6–22)	147	43 (35–51)	24 (17–33)
	B	96	18 (11–27)	2 (0–9)	149	5 (2–9)	4 (2–10)
	C	60	20 (12–32)	18 (11–30)	100	8 (4–15)	6 (3–12)
	D	60	12 (6–22)	18 (11–30)	100	6 (3–12)	6 (3–12)
T3	A	97	7 (4–14)	0 (0–4)	150	2 (1–6)	0 (0–2)
	B	97	7 (4–14)	0 (0–4)	150	2 (1–6)	0 (0–2)
	C	97	7 (4–14)	0 (0–4)	150	3 (1–7)	1 (0–4)
	D	97	7 (4–14)	0 (0–4)	150	3 (1–7)	0 (0–2)
T4	A	97	13 (8–22)	6 (3–13)	150	9 (6–15)	7 (4–12)
	B	97	8 (4–15)	0 (0–4)	150	1 (0–4)	0 (0–2)
	C	97	8 (4–15)	9 (5–17)	150	5 (3–10)	24 (18–31)
	D	97	14 (9–23)	36 (27–46)	150	5 (2–9)	8 (5–13)

A flip is a change of classification at FFR 0.80; *n* is its denominator. Passes and wrong means a territory-perfusion residual below 10% and $|\Delta\text{FFR}| > 0.05$, over models with a defined residual, which reduces *n* for T2 to 60 under B in the discrete bed and to 100 under A and B in the leaky bed. T2 was not applicable to one instance per bed; under A it removed every bed node in 36 discrete and 2 leaky instances, whereas under B the new vessel end carries outflow. Wilson 95% intervals in parentheses. The expected flip rate from repeat invasive measurement is 5.0–6.5% across cells.

vessel breaks (22 reversed, 12 created, $p = 0.24$; mean $\Delta\text{FFR} -0.039$). However, where tuning was defined, only 22% of re-derived topological-error models in the discrete bed (89% in the leaky bed) passed the perfusion check.

B. Passes-and-Wrong Under Tuned Boundary Conditions

Fitting one bed scaling to the clean territory flows (Protocol C) lowered the median perfusion residual of topological-error models from 0.22 under Protocol A to 0.14 in the discrete bed and from 0.11 to 0.02 in the leaky bed, on the same instances. It reversed more topological flips than it created relative to Protocol A (significant for leaky vessel breaks, $p < 0.001$) and changed at most two topological-error decisions per cell relative to Protocol B ($p \geq 0.5$). Protocol D matched the territory flows within 1% in all but two models (Supplementary Material) and corrected most of the topological error: the median $|\Delta\text{FFR}|$ of topological-error models fell to 0.005 in both beds, and their flip rate to 8% (discrete) and 6% (leaky).

Tuning did not remove the error from every model (Fig. 3). After Protocol C, 20 of 104 topological-error models (19%, 13–28%) in the discrete bed and 14 of 171 (8%, 5–13%) in the leaky bed passed the perfusion check while their FFR was materially wrong (more than 0.05 from the clean value); after Protocol D, 21 of 104 (20%, 14–29%) and 9 of 171 (5%, 3–10%) did. On the same instances the proportions were 2% and 6% under Protocol B and 13% and 22% under Protocol A. Tuning raised these proportions relative to B in the discrete bed (McNemar $p < 0.001$ for C and D) but not in the leaky bed ($p \geq 0.12$); relative to A, Protocol C did not change them in the discrete bed ($p = 0.07$) and lowered them in the leaky bed ($p < 0.001$). Models with correct anatomy, tuned to perfusion targets carrying simulated physiological noise (the second floor), passed while materially wrong in 3.1% (2.4–4.0%) and 3.9% (3.2–4.6%) of draws, and their flip rates were

6.2% and 7.2%. Looser thresholds of 13% and 16% raised the proportions (Supplementary Material).

Tuning also turned caliber errors into passes-and-wrong. With re-derived boundary conditions no caliber-error model passed while materially wrong in either bed. After tuning, 5% (C) and 18% (D) did so in the discrete bed and 12% and 4% in the leaky bed (McNemar against B, $p \leq 0.004$ in all four comparisons). Almost all were taper models in which tuning lowered FFR: forcing the clean flow through a uniformly narrowed lumen raises its pressure drop (Table I).

C. Side-Branch Loss Under Fixed and Tuned Resistance

A missed side branch raised the computed FFR. In the 44 (discrete) and 71 (leaky) instances in which tuning was defined, it raised FFR by a median of 0.095 and 0.047 with fixed boundary conditions. Protocol C reduced the shift to 0.077 and 0.015 (Wilcoxon signed-rank, $p < 0.001$ in both beds; Fig. S3 in the Supplementary Material), and Protocol D to 0.000 and 0.004, but 10 of 44 and 3 of 71 instances remained above 0.05 after Protocol D.

D. Three-Dimensional Case Study

The 3D case agreed with the reduced-order result in direction, and prescribed flows removed the error in both. With the clean tree's resistances at the outlets, removal of the side branch raised the 3D FFR at the measurement point from 0.870 to 0.944 ($\Delta\text{FFR} = +0.074$). With the clean territory flows prescribed at the outlets (the limit of Protocol D), the same error changed FFR by -0.0007 (0.892 in both geometries), so in this instance per-territory tuning removed the error. The reduced-order counterparts gave $+0.081$ and -0.0005 for the same two conditions, and their baseline FFR was within 0.011 of the 3D values (Fig. 4). In the cohort's reduced-order model of the same instance (discrete bed, reduced-order

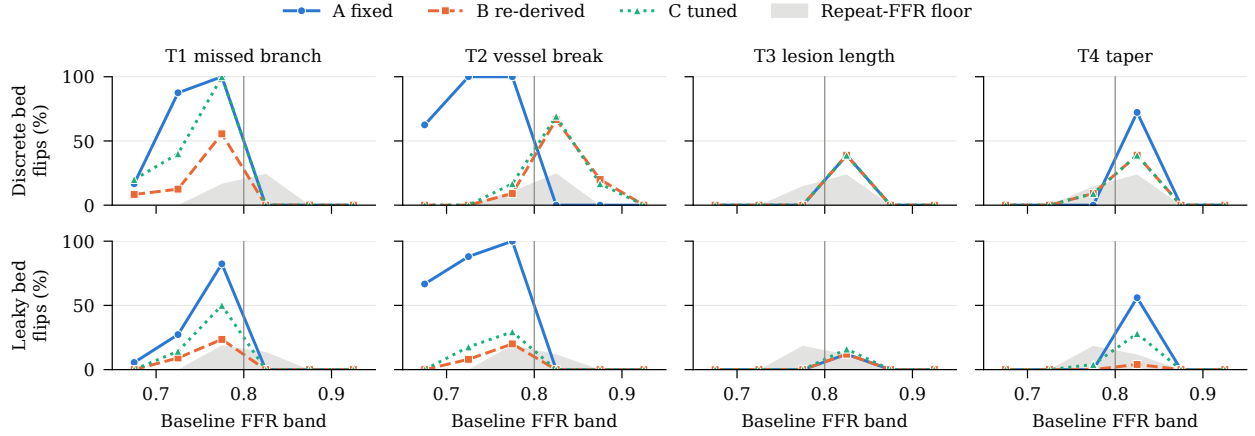


Fig. 2. Flip rate (proportion of models whose classification at FFR 0.80 changed) by baseline FFR band (0.05 wide, plotted at the band center; bands with fewer than three models omitted), error type, protocol (A–C; D in Table I) and bed structure. The shaded area is the flip rate expected from repeat invasive FFR alone (standard deviation 0.018).

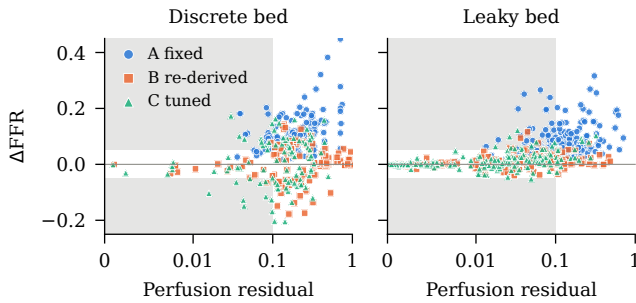


Fig. 3. Change in FFR against territory-perfusion residual for missed-branch and vessel-break models under Protocols A–C (Protocol D residuals lie below 0.01). Each point is one model. The shaded areas hold models that pass the perfusion check (residual < 0.10) while materially wrong ($|\Delta\text{FFR}| > 0.05$). The horizontal axis is linear below 0.01 and logarithmic above.

radius), the error gave $+0.127$ under Protocol A and -0.0007 under Protocol D, whereas the single scaling of Protocol C left $+0.107$ with a residual of 0.30, which fails the check. Two caveats apply. The meshed lumen was wider than the radius used by the reduced-order model (median difference 0.14 mm; throat area-equivalent radius 0.276 against 0.231 mm), and on the reduced-order radius the baseline was 0.761. A finer throat zone (12.5 μm) changed the baseline FFR by at most 0.0021, against the error effect of 0.074 (Supplementary Material).

E. Hyperemic Demand

With k doubled and tripled, the lesion insertion, cohort selection and all four protocols were repeated (Supplementary Material). Fewer trees met the discrete-bed criteria (47 and 17 instances; the discrete arm is reported at twice the demand only). With fixed boundary conditions, topological errors changed 33–50% of decisions against 1–5% for caliber errors; tuning reduced topological flips to 3–20%; and after tuning, 3–20% of topological-error and 1–14% of caliber-error models passed while materially wrong, against 2–5% and 0% with re-derived boundary conditions. These directions held

in both beds at twice and in the leaky bed at three times the demand; at twice the demand the excess over re-derived boundary conditions was significant in three of four caliber comparisons and in no topological one.

IV. DISCUSSION

A. What a Perfusion Check Shows After Tuning

With fixed boundary conditions, the decision risk of segmentation lay mainly in the branching structure of the lumen: a topological error changed the decision at 0.80 three to six times as often as a caliber error of inter-observer magnitude, whose flip rates were of the order of repeat invasive measurement. This ranking holds for the magnitudes studied.

Re-deriving or tuning the bed reduced topological flips from 18–43% to 2–20% per error type. Per-territory tuning restored the flow that the missing branch had carried through the stenosis, in the cohort and in the 3D case, by construction of its targets.

After tuning, a passing perfusion check did not show that the FFR was correct. Per-territory tuning passes almost every model by design, and in the discrete bed one in five tuned topological-error models passed while materially wrong under either form of tuning. After one global scaling, about half of the models that passed were materially wrong. In both beds tuning turned caliber errors into passing models with a wrong FFR, which re-derivation did not, because restoring the clean flow through a narrowed lumen enlarges its pressure drop. The check measured agreement with the fitted targets, not the correctness of the model.

Bed structure set the magnitude. The leaky bed damped topological errors, and their excess of passes-and-wrong over re-derived boundary conditions held only in the discrete bed.

B. Relation to Published Studies

Published sensitivity analyses perturb lumen caliber at fixed branching, and comparisons of geometry with boundary conditions rank them on a continuous output [14]–[16]. Our

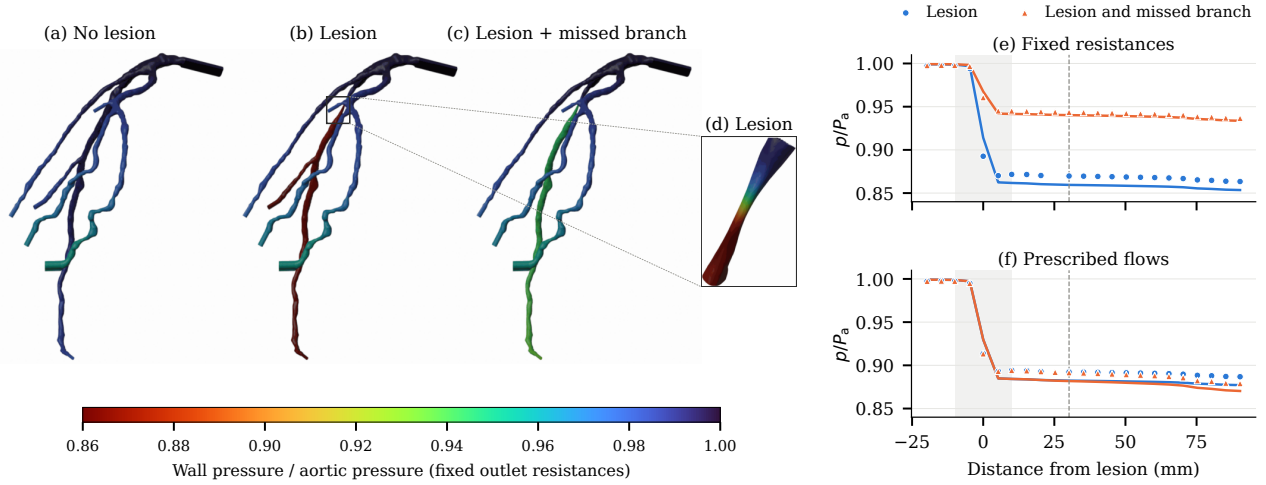


Fig. 4. Three-dimensional case study (proximal LAD, 80% DS). (a)–(c) Wall pressure relative to aortic pressure with the clean tree’s outlet resistances, without the lesion, with the lesion, and with the lesion and a missed side branch. The lesion lowers the downstream pressure (red); the missed branch raises it again (green), a shift that would move a decision near 0.80. (d) Close-up of the lesion boxed in (b). (e), (f) Pressure along the artery with fixed resistances and with prescribed clean territory flows. Markers show 3D cross-section averages and lines the reduced-order models on the meshed radius with the same outlet conditions. The shaded band is the lesion and the dashed line the measurement point.

results add that the ranking depends on the error type and the boundary-condition protocol.

Two earlier findings that appeared to disagree are both consistent with our results: a side-branch effect of 13–15.5% under fixed outlet resistance [20], and little change in diagnostic accuracy after a cohort-wide outlet resistance was recalibrated [21]. Both appear in our cohort, the first with fixed resistance and the second after re-derivation or tuning, so the boundary-condition protocol alone can produce both behaviors.

Vessel breaks occur in the output of current segmentation methods despite high overlap scores [18], and a missing side branch removes few voxels, so the error types that carried the decision risk here are the ones that the usual overlap metrics do not measure, although topology-aware measures such as cIDice exist [32].

C. Implications and Practical Use

Agreement with a fitted quantity does not by itself make a model credible. A coronary digital twin calibrated to perfusion also requires verification, uncertainty quantification and applicability [33]. A missed branch that survived tuning raised FFR, biasing the decision toward deferring treatment, whereas a tuned taper error lowered it, biasing the decision toward treatment.

For a user who computes FFR from an image, the results suggest four considerations. First, check that the side branches beyond the lesion are present and that no vessel ends early, because overlap scores do not reveal these errors. Second, if the boundary conditions are tuned to perfusion, check the lumen caliber as well: diameter errors of inter-observer size flipped decisions at rates near repeat invasive measurement (the taper up to 14%), but after tuning they produced passing models with a wrong FFR. Third, record the perfusion mismatch before tuning, which erases it: in the discrete bed only 22% of re-derived topological-error models passed, although in the leaky bed most did. Fourth, take most care near 0.80,

where decision changes concentrated. These considerations follow from the results but are not validated decision rules.

D. Limitations

The inserted stenoses are idealized, and the reduced-order model is steady, without autoregulation or heart-rate effects. The missed-branch and vessel-break magnitudes are design choices, and the caliber magnitudes come from inter-observer statistics that likely overstate agreement. Protocols C and D were undefined for most right coronary instances with a topological error, so their results for these errors describe the left coronary tree. Hyperemic demand followed the segmented inlet radius and was below thermodilution values. The 3D analysis covers one instance, whose meshed lumen was wider than the reduced-order radius; this offset moved the baseline across 0.80 (0.761 against 0.870) and changed the size of the error effect (+0.127 against +0.081 on the meshed radius), but not its direction or its removal by prescribed flows. No invasive FFR was available, so the noise floor was modeled from published repeat measurements. Errors were applied one at a time in one dataset. Tuning targets were error-free clean-model flows (noise entered only the simulated floor).

V. CONCLUSION

This study proposes a controlled test that introduces one segmentation error at a time into a coronary flow model and judges its effect by the treatment decision at FFR 0.80. Applied to 150 inserted stenoses in 108 trees, it showed that a model that matches perfusion is not necessarily a correct one. With fixed boundary conditions, missed branches and vessel breaks changed the decision three to six times as often as caliber errors of inter-observer size. Tuning the boundary conditions to perfusion reduced these changes without making the models correct: in the discrete bed, up to one in five tuned topological-error models (5–8% in the leaky bed) passed a perfusion

check stricter than measurement repeatability while their FFR was materially wrong, against 3–4% for correct anatomy. The results point to where safeguards are likely to matter most: a check of the branching near the lesion, which overlap scores miss, and a record of the perfusion mismatch before tuning, which tuning erases. The same test can be adapted to other segmentation and boundary-condition pipelines, including coronary digital twins calibrated to perfusion.

DATA AND CODE AVAILABILITY

The data that support the findings of this study were derived from resources available in the public domain: ImageCAS [22] and ImageCAS-X [23] (CC BY 4.0). The code is available from the corresponding author upon reasonable request.

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